MSG No. 191.240 $P6/mm'm'$ [Type III, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `6/mm'm'`

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: `1b`, site symmetry: `6/mm'm'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: `2c`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,3,5,8,10,12,16,17,18,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[2,4,6,7,9,11,13,14,15,22,23,24]

Table 4: Wyckoff site: `2d`, site symmetry: `-6m'2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,3,5,8,10,12,16,17,18,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[2,4,6,7,9,11,13,14,15,22,23,24]

Table 5: Wyckoff site: `2e`, site symmetry: `6m'm'`

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,4,5,6,19,20,21,22,23,24]
2	$[0, 0, -z]$	[7,8,9,10,11,12,13,14,15,16,17,18]

Table 6: Wyckoff site: `3f`, site symmetry: `mm'm'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,7,10,13,17,19,23]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,5,8,11,15,16,21,22]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,6,9,12,14,18,20,24]

Table 7: Wyckoff site: `3g`, site symmetry: `mm'm'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,4,7,10,13,17,19,23]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,5,8,11,15,16,21,22]

3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,6,9,12,14,18,20,24]`

Table 8: Wyckoff site: `4h`, site symmetry: `3m'`.

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,3,5,19,20,21]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[2,4,6,22,23,24]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,9,11,13,14,15]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[8,10,12,16,17,18]`

Table 9: Wyckoff site: `6i`, site symmetry: `2m'm'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4,19,23]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[2,5,21,22]`
3	$\left[0,\ \frac{1}{2},\ z\right]$	`[3,6,20,24]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,10,13,17]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[8,11,15,16]`
6	$\left[0,\ \frac{1}{2},\ -z\right]$	`[9,12,14,18]`

Table 10: Wyckoff site: `6j`, site symmetry: `m2'm'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,10,13,23]`
2	$[x,\ x,\ 0]$	`[2,11,16,21]`
3	$[0,\ x,\ 0]$	`[3,12,14,24]`
4	$[-x,\ 0,\ 0]$	`[4,7,17,19]`
5	$[-x,\ -x,\ 0]$	`[5,8,15,22]`
6	$[0,\ -x,\ 0]$	`[6,9,18,20]`

Table 11: Wyckoff site: `6k`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,10,13,23]`
2	$\left[x,\ x,\ \frac{1}{2}\right]$	`[2,11,16,21]`
3	$\left[0,\ x,\ \frac{1}{2}\right]$	`[3,12,14,24]`
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,7,17,19]`
5	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[5,8,15,22]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,9,18,20]`

Table 12: Wyckoff site: `6l`, site symmetry: `mm'2'`

No.	position	mapping
1	$[x,\ 2x,\ 0]$	`[1,10,17,19]`
2	$[-x,\ x,\ 0]$	`[2,11,15,22]`
3	$[-2x,\ -x,\ 0]$	`[3,12,18,20]`
4	$[-x,\ -2x,\ 0]$	`[4,7,13,23]`
5	$[x,\ -x,\ 0]$	`[5,8,16,21]`
6	$[2x,\ x,\ 0]$	`[6,9,14,24]`

Table 13: Wyckoff site: `6m`, site symmetry: `mm'2'`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1,10,17,19]`
2	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[2,11,15,22]`
3	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[3,12,18,20]`
4	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4,7,13,23]`
5	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[5,8,16,21]`
6	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[6,9,14,24]`

Table 14: Wyckoff site: `12n`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,23]`
2	$[x,\ x,\ z]$	`[2,21]`
3	$[0,\ x,\ z]$	`[3,24]`
4	$[-x,\ 0,\ z]$	`[4,19]`
5	$[-x,\ -x,\ z]$	`[5,22]`
6	$[0,\ -x,\ z]$	`[6,20]`
7	$[-x,\ 0,\ -z]$	`[7,17]`
8	$[-x,\ -x,\ -z]$	`[8,15]`
9	$[0,\ -x,\ -z]$	`[9,18]`
10	$[x,\ 0,\ -z]$	`[10,13]`
11	$[x,\ x,\ -z]$	`[11,16]`
12	$[0,\ x,\ -z]$	`[12,14]`

Table 15: Wyckoff site: `12o`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 2x,\ z]$	`[1,19]`
2	$[-x,\ x,\ z]$	`[2,22]`
3	$[-2x,\ -x,\ z]$	`[3,20]`
4	$[-x,\ -2x,\ z]$	`[4,23]`
5	$[x,\ -x,\ z]$	`[5,21]`
6	$[2x,\ x,\ z]$	`[6,24]`
7	$[-x,\ -2x,\ -z]$	`[7,13]`

continued ...

Table 15

No.	position	mapping
8	$[x, -x, -z]$	[8,16]
9	$[2x, x, -z]$	[9,14]
10	$[x, 2x, -z]$	[10,17]
11	$[-x, x, -z]$	[11,15]
12	$[-2x, -x, -z]$	[12,18]

Table 16: Wyckoff site: 12p, site symmetry: m..

No.	position	mapping
1	$[x, y, 0]$	[1,10]
2	$[x-y, x, 0]$	[2,11]
3	$[-y, x-y, 0]$	[3,12]
4	$[-x, -y, 0]$	[4,7]
5	$[-x+y, -x, 0]$	[5,8]
6	$[y, -x+y, 0]$	[6,9]
7	$[x-y, -y, 0]$	[13,23]
8	$[y, x, 0]$	[14,24]
9	$[-x, -x+y, 0]$	[15,22]
10	$[x, x-y, 0]$	[16,21]
11	$[-x+y, y, 0]$	[17,19]
12	$[-y, -x, 0]$	[18,20]

Table 17: Wyckoff site: 12q, site symmetry: m..

No.	position	mapping
1	$\left[x, y, \frac{1}{2}\right]$	[1,10]
2	$\left[x-y, x, \frac{1}{2}\right]$	[2,11]
3	$\left[-y, x-y, \frac{1}{2}\right]$	[3,12]
4	$\left[-x, -y, \frac{1}{2}\right]$	[4,7]
5	$\left[-x+y, -x, \frac{1}{2}\right]$	[5,8]
6	$\left[y, -x+y, \frac{1}{2}\right]$	[6,9]
7	$\left[x-y, -y, \frac{1}{2}\right]$	[13,23]
8	$\left[y, x, \frac{1}{2}\right]$	[14,24]
9	$\left[-x, -x+y, \frac{1}{2}\right]$	[15,22]
10	$\left[x, x-y, \frac{1}{2}\right]$	[16,21]
11	$\left[-x+y, y, \frac{1}{2}\right]$	[17,19]
12	$\left[-y, -x, \frac{1}{2}\right]$	[18,20]

Table 18: Wyckoff site: 24r, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[x-y, x, z]$	[2]

continued ...

Table 18

No.	position	mapping
3	$[-y,\, x-y,\, z]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[-x+y,\, -x,\, z]$	[5]
6	$[y,\, -x+y,\, z]$	[6]
7	$[-x,\, -y,\, -z]$	[7]
8	$[-x+y,\, -x,\, -z]$	[8]
9	$[y,\, -x+y,\, -z]$	[9]
10	$[x,\, y,\, -z]$	[10]
11	$[x-y,\, x,\, -z]$	[11]
12	$[-y,\, x-y,\, -z]$	[12]
13	$[x-y,\, -y,\, -z]$	[13]
14	$[y,\, x,\, -z]$	[14]
15	$[-x,\, -x+y,\, -z]$	[15]
16	$[x,\, x-y,\, -z]$	[16]
17	$[-x+y,\, y,\, -z]$	[17]
18	$[-y,\, -x,\, -z]$	[18]
19	$[-x+y,\, y,\, z]$	[19]
20	$[-y,\, -x,\, z]$	[20]
21	$[x,\, x-y,\, z]$	[21]
22	$[-x,\, -x+y,\, z]$	[22]
23	$[x-y,\, -y,\, z]$	[23]
24	$[y,\, x,\, z]$	[24]